

EECS 464: Hands-on-Robotics Project 2 Report

Purple Team

Luyang Hu, Yichen Zhu, Yuesheng Ren, George Deng

December 2022



Contents

1	Background	1
1.1	Project Task Specification	1
1.1.1	Problem Statement	1
1.1.2	Pass/Fail Requirement	1
1.1.3	Performance Goal	1
1.2	Possible Designs	1
1.2.1	H Frame + Movable Base	1
1.2.2	5 DoF Robotic Arm with Counter Weights	2
1.2.3	H Frame + Scissor Structure Base	2
1.3	Final Design	2
1.3.1	Mechanical Design	2
1.3.2	Software Design	3
2	Results	4
3	Discussion	4
3.1	Overall Analysis	4
3.2	Our Design	4
3.2.1	Hardware Analysis	5
3.2.2	Software Analysis	5
3.3	Other Designs	5
3.3.1	Team Maize	5
3.3.2	Team Blue	6
3.3.3	Team Red	6
3.3.4	Team Green	6
3.4	Conclusion	7

1 Background

1.1 Project Task Specification

1.1.1 Problem Statement

This report describes the details regarding Project 2 in the Hands-On Robotics (HRB) class during Fall 2022. In this project, each team was tasked to build a robot that was capable of drawing on a piece of paper that would be placed in a 33 x 33 x 33 cm cube workspace. The paper's normal $[x,y,z]$ would be in the direction of one of $[1,0,0]$, $[0,1,0]$, $[0,0,1]$, $[1,0,1]$, $[1,1,0]$, $[0,1,1]$, or $[1,1,1]$. On the day of the competition (P-day), each team had a ten-minute calibration period, during which each team could manually guide the pen in any position and/or orientation in the work area or manually move their robot. Afterwards, each team got ten minutes of drawing time, and the instructor announced the (X, Y) target coordinates and the scale S. The goal was for the robot to draw a square at the target location with the specified scale.

1.1.2 Pass/Fail Requirement

The pass/fail requirements for Project 2 were as follows:

1. Draw a closed shape on paper with four identifiable corners and at least two orientations.
2. The shortest "side" of the shape must be at least 5 cm.

1.1.3 Performance Goal

Each team will be scored on its run with the best performance. The score for the drawing will be the RMS distance between the markings drawn and a theoretical perfect square at the target location and scale.

1.2 Possible Designs

1.2.1 H Frame + Movable Base

A proposed design of the **H frame** is shown in Fig 1. In the H Frame design, only two motors were needed. This design allowed drawing simplified control by decoupling the x and y axes on the drawing plane. However there were some limitations when using it, it has to be parallel to the drawing surface and the structure is complex as it required the team to mount eight pulleys in total. To move the H frame onto the page, this design added a movable base and an arm to hold the frame. The movable base was inspired by [Green Team's design from Project 1](#), as shown in Fig 2. With this base, the location of the whole system could be handled by controlling the motors and it could also decrease the complexity in controlling the arm.

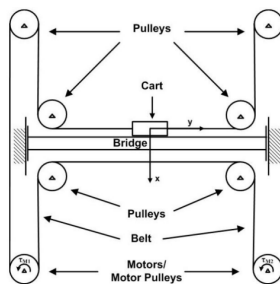


Figure 1: Mechanical structure of H Frame

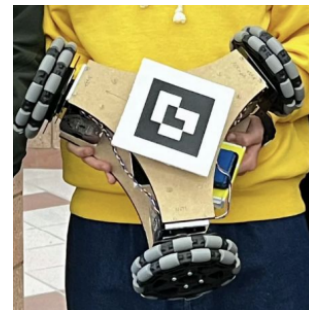


Figure 2: Mechanical structure of movable base

1.2.2 5 DoF Robotic Arm with Counter Weights

The second design was a five degrees of freedom robotic arm and is shown in Fig 3. In this design, only five degrees of freedom were needed because we would remove the sixth axis which was used to rotate the pen. This design allowed our robot to reach every point in the work zone. At the same time, this design also produced large loads on the motors and would be more complex to control.



Figure 3: Mechanical structure of 6 DoF robotic arm

1.2.3 H Frame + Scissor Structure Base

The third design was an H-Frame buggy with a scissor mechanism and is shown in Fig 4. This design was inspired by the 2017 blue team. However, the 2017 blue team's design could not reach the page if placed horizontally at the top of the 33cm x 33cm x 33cm box. We decided to add a scissor structure on the base to solve this problem. In this design, motors 1 and 2 would control the two scissor structures to make the base go up and down. Motor 3 would rotate the H-frame. Meanwhile, Motor 4 and 5 controlled the H-frame so the robot could draw a 2d shape in any orientation.

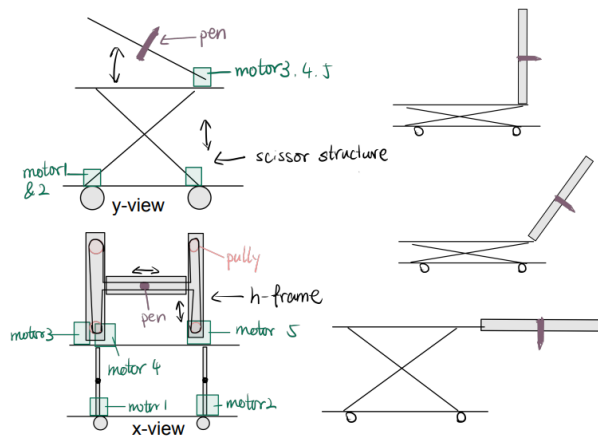


Figure 4: Mechanical structure of 5 DoF robotic arm

1.3 Final Design

1.3.1 Mechanical Design

The final mechanical design consisted of a movable base with a three degrees of freedom arm. We decided to go with a three omni-wheel design for the movable base. We laser cut a fiberboard base with two sets of radial slots whose widths were 4mm, which were used in conjunction with 3D-printed supports to increase the stability of the robotic arm. The radial slots were long enough to allow the arm to reach every possible point in the 33cm x 33cm x 33cm workspace. This mechanism can be seen in figure ?? and ??.

For the actual arm, we added 325 grams of counterweights to the second link to reduce the load on the 3D-printed pen holder motors. We utilized the winter 2022 Maize team's [3D-printed pen holder](#), which pressed the pen onto the paper with rubber bands. The second link consisted of a double arm design with two foam core structures, one to each side of the motor. This design was introduced to increase arm stability. The final robot design can be seen in Fig 7.



Figure 5: Image of 3D printed supports

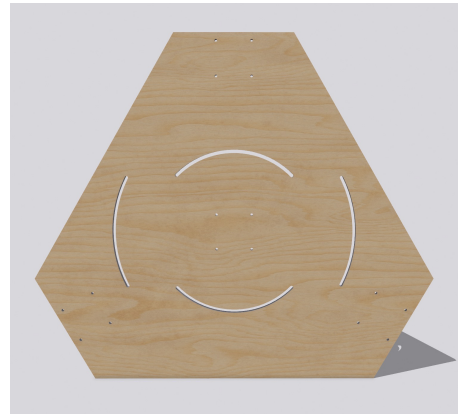


Figure 6: Image of laser cut base

1.3.2 Software Design

For our software design, we built our software based on [Maize 2022W](#) . Using their forward and inverse kinematics, we built our own version of the keyframe.py program in pyckbot to make the tip of the pen interpolate the motion from one point to the next utilizing inverse kinematics. To better estimate the current position of the robot, we read the position of the motors and used direct kinematics to tell exactly where we were. Then we fed the position back into the inverse kinematics to determine the next step. We communicated with the motors at 7Hz to prevent an overflow on the data bus. During calibration, we recorded multiple points at each side of our desired square. We also extended each side by about 3cm to ensure a sharp corner. We used continuous mode for each motor and set the motor speed to 3 to counter gravity and static friction. We also implemented [Green team's 2022 Project 1](#) buggy control algorithm to control the location of our robot arm. Using keyboard control, we were able to move the robot base and the arm, record the position of the robot arm, and replay the motion.

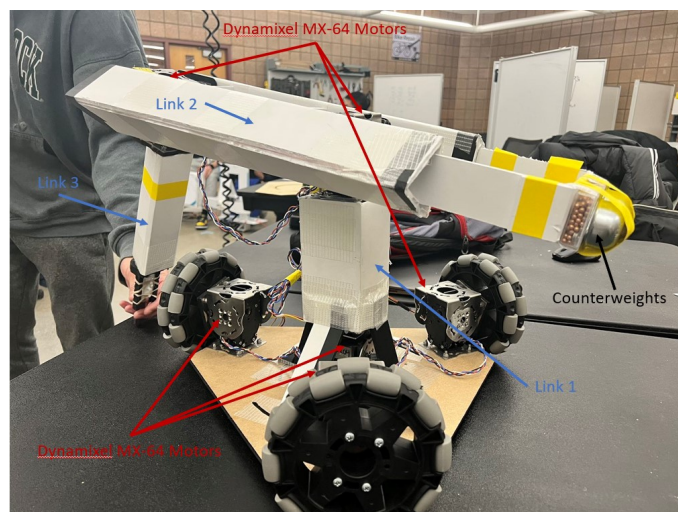


Figure 7: Mechanical structure of final robot

2 Results

The results are shown in Table 1. All raw, marked and scored drawings can be found [here](#).

Team	Paper Normal [x, y, z]	(1, 1, 0)	(1, 1, 0)	(0, 0, 1)	(0, 0, 1)	(0, 1, 0)	Extra	
	Paper Height (cm)	0	0	33	33	0		
	Target Square Center (X,Y) (cm)	(10, 12)	(15, 20)	(14, 12)	(14, 12)	(15, 15)		
	Target Square Scale S (cm)	6	10	8	4	10		
Maize	RMS	NA	111.23	354.36	338.43	150.28	-	-
Blue		NA	NA	NA	NA	NA	305.35	198.49
Red		93.30	NA	NA	NA	NA	107.15	-
Green		31.71	NA	76.37	124.18	20.41	-	-
Purple		64.16	51.81	47.33	28.48	378.61	-	-

Table 1: Project 2 Results. Paper height was the Z component of the origin of the paper coordinate frame in the NA = failed to draw; Extra Round only for groups that did not meet the PF criteria in previous rounds.

Purple team was the only group that are able to draw all give orientation. Green and Maize team were also close to it, but generally maize team had large error in terms of drawing it in desired position. Blue team had trouble drawing a closed shape. From there drawing, we can see that their pen was constantly lifted up. Red team drew almost perfect on the first attempt. However, because of some software issue, they cannot drew in other orientations.

3 Discussion

3.1 Overall Analysis

All teams successfully completed the pass/fail criteria. The Purple, Maize, and Green teams were the most consistent and were able to draw a closed shape with four distinct corners in most orientations. However, no team was able to consistently draw a square in the correct location with the correct scale. We believe this can largely be split into to two reasons: Hardware failures and Software failures.

In terms of hardware failures, on P-Day we observed multiple teams struggling to maintain contact between the sharpie and the page. Often, we saw unstable motion leading to less consistent contact with the page and occasional separation from the page. On the other side, we also observed a few teams pushing against the paper with excess pressure. The pen would not move until it jumped off the page and skipped to the next point.

In terms of software, we observed many robots with potentially underdeveloped software. We observed many trials where robots would perform unintended actions. There were multiple cases where the robots would attempt to draw off of the page or move the pen away from the page.

3.2 Our Design

Overall our design drew a closed shape with four identifiable corners in every orientation. In the vertical orientation, our robot required multiple passes to ensure the square was closed. Our best RMS error score was 28.48. Unfortunately, our software for drawing in the correct location at the correct scale did not work. Thus, all square locations and sizes were predetermined during the calibration phase.

3.2.1 Hardware Analysis

Overall our hardware was reliably able to draw on all orientations. One of the biggest benefits of our design was the movable base. During calibration, we were able to precisely choose the best location for our robotic arm. This not only reduced the number of orientations the arm had to face but also allowed us to place the arm at the most optimal distance to the page no matter where it was. We believe the 3D printed supports at the base of our first link and the double-sided design of the second link increased the stability of our robot during its motion.

While operating our robot, we occasionally saw the wheels slip while pushing against the page. We believe this could be improved by changing the base design to two regular wheels with differential drive. We believe the wheels would get more friction with two regular wheels as opposed to our three omni-wheel designs. The omni-wheel design was chosen out of convenience.

3.2.2 Software Analysis

Unfortunately, we were unable to get our software to draw a square in the correct location with the correct scale. We believe this was largely due to our calibration protocol which was supposed to run along the edges of the paper to calibrate the paper location in terms of motor encoder counts. During testing, our calibration was inconsistent and only produced usable results from horizontal orientations at a height of 5 cm to 20cm. However, we noticed our calibration protocol could reliably draw between any points recorded. During the competition, we used the calibration protocol to draw every square and simply guessed the location and size of the square beforehand. We believe with more time dedicated to debugging our calibration software our robot's performance could greatly improve.

While calibrating the robot, we were also required to physically move the arm to record the points, meaning the motors were deactivated. We believe implementing a motion script to move the motors to the calibration points would have increased our calibration's precision. We did not implement this due to time constraints during P-day.

3.3 Other Designs

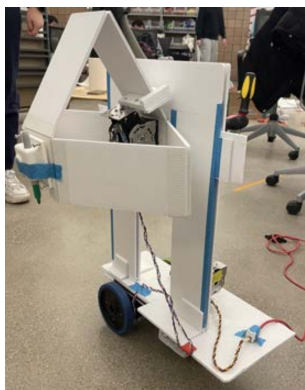


Figure 8: Maize Team's Robot

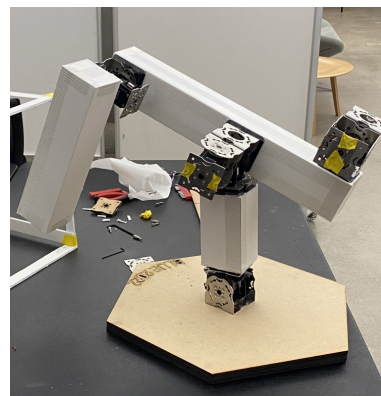


Figure 9: Blue Team's Robot

3.3.1 Team Maize

The Maize team's robot shown in figure 8 was able to draw a closed shape with four corners in four out of five configurations. Their best RMS error among these trials was 150.28. They have a unique design that allows the pen to move in all three directions independently. Their robot performed well when drawing horizontally and

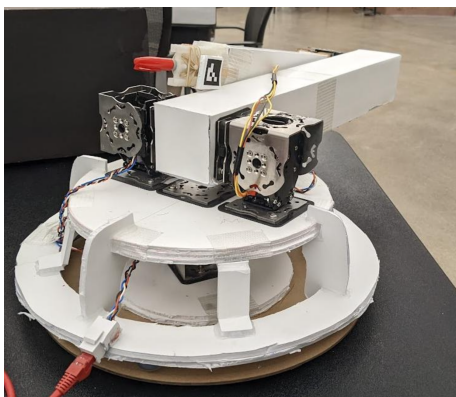


Figure 10: Green Team's Robot



Figure 11: Red Team's Robot

vertically. However, due to their unique structure, the shape did not always have sharp corners when drawing on 45-degree inclined surfaces. They also failed to put squares at the designated locations with the correct size. We believe this is due to flaws in their software that failed to accurately control each axis.

3.3.2 Team Blue

The blue team's robot shown in figure 9 was able to draw closed shapes on the last two out of seven configurations. Their best RMS score was 198.49. Their mechanical design was a three-degree-of-freedom robotic arm. Most of their failed trails had square-like shapes but were not closed. We believe this is partly because they purely used keyframe.py to do the software control of their robot. Keyframe.py only records the position of the motors between each point. The program did not use proper inverse kinematics to interpolate between steps resulting in only dots on their paper. Mechanically, their robot had used extra motors counterweights but do not have support at the base.

3.3.3 Team Red

The red team's robot shown in figure 11 was able to draw a closed shape with four corners in two out of six configurations. Their best RMS error among these trials was 93.30. Overall, the red team struggled to make consistent contact with the paper. Four out of six trails from the red team resulted in no or minimal markings. During these attempts, we observed their arm consistently move up and remove the pen from the paper. We believe this may be due to a bug in their software or poor inverse kinematics. We believe with improved software their robot would have much more consistently drawn on the page. Mechanically their robot was well supported with proper counterweights. They implemented supports similar to ours at the base of their robotic arm.

3.3.4 Team Green

The green team's robot shown in figure 10 was able to draw a closed shape with four corners in every attempt except one. Their best RMS error was 20.41, the best of the day. Overall, the green team's mechanics were consistently able to move the pen along the page without losing connection to the page. We believe the foam core "skirt" around the base of their arm improved their arm's stability. They also utilized two motors connected to the second link to increase the amount of torque provided. This allowed them to move their arm without other physical gravity compensation.

Unfortunately, their software was not as reliable. We observed the green team struggle to achieve a strong calibration. In their second trial, the arm started to draw off of the page. After this they decided to switch gears

and simply record motor encoder counts and replay them using their "keyframes" script. This meant they were simply guessing at the location and scale of the square. We believe improving their calibration methods would lead to large performance gains.

3.4 Conclusion

Overall, our robot performed well. We achieved the second lowest RMS error of 28.48 and were the only team to draw a closed shape with four corners on every attempt. Unfortunately, we were not able to consistently draw in the correct location and scale. However, with a few improvements to the calibration program, we believe we could improve our score.

This project taught us about many of the real world challenges associated with designing a robotic arm. In particular, we learned about many practical concerns such as motor limits, arm stability, and gravity compensation. This project also taught us a lot about the importance of software simulation and integration issues. If we were to do this project again we would attempt to more thoroughly develop our software and finish our mechanical design earlier.